



Django MVT Architecture (Lecture #2)

Deep-Dive: Model, View, Template | 6-Step Workflow | MVT vs MVC Comparison

Master Class Notes • Mohit Decodes Playlist #2 • Hinglish Quick Revision

🎯 Django Ki Backbone (Personal Context & Link):

Django me MVT Architecture project ka *Backbone* hota hai jo request se lekar response tak ke pure raste ko guide karta hai.

- **Model (SQL / Data Layer):** Yahan aapki SQL knowledge kaam aati hai. Model table ka blueprint hai jo database schema aur data management sambhalta hai.
- **View (Python Logic):** Yahan aapka Python code chalega (Functions / Classes) jo faisle leta hai ki user ko kya data dikhana hai.
- **Template (Presentation / UI):** HTML + Django Template Language (DTL) jo data ko user-friendly style me render karti hai.



1. MVT Ke 3 Core Pillars (Components)

1. MODEL (M)

Database Representation:

- Data ko store aur manage karta hai.
- Database tables ko Python classes ke roop me define karta hai.
- SQL queries ko internally execute karke data lata hai.

2. VIEW (V)

Business Logic (Brain):

- User request ko receive karta hai.
- Model se interact karke data fetch karta hai.
- Data processing karke Template ko pass karta hai.

3. TEMPLATE (T)

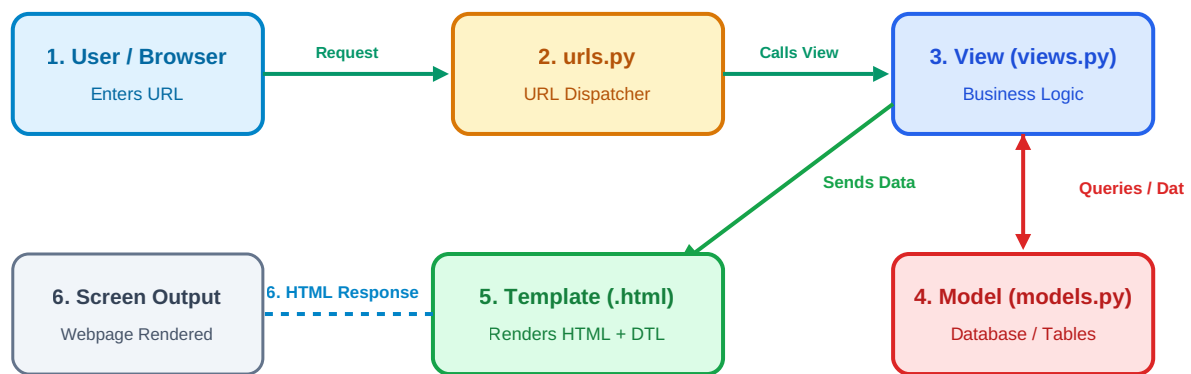
Presentation UI:

- **HTML + DTL** (Django Template Language).
- View se mile dynamic data ko visual format me render karta hai.
- User ko clean webpage ke roop me dikhta hai.



2. MVT 6-Step Workflow (Request-to-Response Journey)

Jab koi user browser me click karta hai, toh Django ke andar 6 steps me execution hota hai:



Step	Component	Parde Ke Peeche Kya Hota Hai?
Step 1	User / Browser	User browser open karke koi URL paste/click karta hai (e.g. <code>/products/</code>) aur Enter press karta hai.
Step 2	URL Dispatcher (<code>urls.py</code>)	Django ka traffic police! Ye check karta hai ki is URL ke liye kaun sa Python View function/class trigger karni hai.
Step 3	View Function / Class	View user ki request ko samjhta hai aur Model ke pass jakar bolta hai: <i>"Bhai, mujhe ye data chahiye!"</i>
Step 4	Model (Database)	Model database check karta hai ki manga hua data available hai ya nahi. Data milte hi View ko return kar deta hai.
Step 5	Template (<code>.html</code>)	View us raw data ko Template ke paas bhejta hai. Template HTML design me dynamic data fill (render) karta hai.
Step 6	HTTP Response (User Screen)	Final formatted HTML page user ke browser par display ho jata hai.

🍕 Hotel / Food Order Analogy (Bachchon Jaisa Simple):

- **Step 1 (User):** Aapne waiter ko menu dekh kar kaha: *"Table #4 ke liye ek Chicken Biryani!"* (URL Hit).
- **Step 2 (URL Dispatcher):** Waiter ne counter pe slip dekhi ki ye order konse kitchen chef ke paas bhejte hain (`urls.py`).
- **Step 3 (View / Chef):** Head Chef (View) ne recipe samjhi aur assistant (Model) ko store room se chawal aur chicken laane ko kaha.
- **Step 4 (Model / Store Room):** Assistant ne pantry (Database) check kiya, raw chicken aur rice nikal kar Chef ko de diya.
- **Step 5 (Template / Plate Decoration):** Chef ne khana banaya aur sundar si ceramic plate (HTML Template) me garnish karke sajaya.
- **Step 6 (Response):** Waiter ne sajaya hua garam khana aapke table par serve kar diya (Page display ho gaya)!



3. MVT vs MVC (Sabse Common Interview Question)

Log aksar MVC (Model-View-Controller) aur MVT me confuse hote hain. Asal me **MVT ek variant (variation) hi hai MVC ka**. Dono me sirf naam ka farq hai:

MVC (Traditional Pattern)	Django MVT (Django Ka Pattern)	Kaam (Role / Responsibility)
Model (M)	Model (M)	Dono me bilkul same hai! Database tables aur data storage ko handle karna.
Controller (C)	View (V)	Jo kaam MVC me <i>Controller</i> karta hai, wahi kaam Django me View karta hai (Business logic & data flow control).
View (V)	Template (T)	Jo kaam MVC me <i>View</i> karta hai, wahi kaam Django me Template karta hai (User ko HTML UI dikhana).



Pro Tip for Interviews:

Agar interviewer pooche: "Django MVC hai ya MVT?"

Perfect Answer: "Sir, Django MVT architecture follow karta hai, jo MVC ka hi ek variant hai. Farq sirf itna hai ki MVC ka Controller yahan Django me 'View' kehlata hai, aur MVC ka View yahan 'Template' kehlata hai. MVC ka Controller role Django ka Framework khud internally manage karta hai through URL routing and Views."



4. Top 5 Interview Quick Revision Points

1. **Django ki backbone kis architecture ko kaha jata hai?** → **MVT** (Model-View-Template) Architecture.
2. **`urls.py` ka kya role hota hai?** → Ye *URL Dispatcher* hai jo incoming URL ko match karke sahi View function ko call karta hai.
3. **Model aur Database ke beech kya rishta hai?** → Model Python class hoti hai jo database table ka blueprint hoti hai aur ORM ke through data fetch/store karti hai.
4. **Template me DTL ka full form kya hai?** → **Django Template Language** (jo HTML me dynamic Python variables `{{ variable }}` aur loops `{% for %}` lagane ki suvidha deta hai).
5. **MVC ke Controller aur MVT ke View me kya similarity hai?** → Dono ka kaam business logic likhna aur request-response flow ko coordinate karna hai.